



DAYANANDA SAGAR COLLEGE OF ENGINEERING
(An Autonomous Institute affiliated to Visvesvaraya Technological University (VTU), Belagavi,
Approved by AICTE and UGC, Accredited by NAAC with 'A' grade & ISO 9001-2015 Certified Institution)
Shavige Malleshwara Hills, Kumaraswamy Layout, Bengaluru - 560 111. India

Department of Information Science and Engineering
(Accredited by NBA Tier 1: 2025-2028)

DSCE ISE ACM STUDENT CHAPTER

Events Conducted in Academic Year 2024-25

Sl. No.	Description	Date
1.	From LLM's to RAG: Enhancing AI Capabilities with External Knowledge	03 rd Oct 2024
2.	Technical Treasure Hunt	07 th Oct 2024
3.	Tech Trek - An 8-hour Open Theme Hackathon	29 th March 2025
4.	CypherQuest 2k25 - 24 Hours ACM Zonal Level Hackathon	25 th & 26 th April 2025
5.	Developers' Conclave: Industry Driven Insights from Industry Insiders	26 th April 2025
6.	Aventus 3.0 - 24 hrs National Level Hackathon	17 th & 18 th of May 2025
7.	AI-assisted Tutoring for CS1	27 th June 2025

DSCE ISE ACM STUDENT CHAPTER

Member List

ACM Faculty Sponsor: Dr.Rajeshwari J (4814426)

ACM Faculty Co-ordinators:

Sl. No.	Name	ACM Membership - ID
1.	Mrs. Shalini K B	6232471
2.	Mr. Bharath B C	5443321

ACM Faculty Members:

Sl. No.	Name	ACM Membership - ID
1.	Dr. Annapurna P Patil	5527307
2.	Dr. Madhura J	3291877
3.	Dr.Vani K A	5127513
4.	Dr.Prathima Mabel J	8819846
5.	Mr.Praveen N	3738349
6.	Dr.Latha A P	8408408
7.	Dr. Bhavani K	8312044

ACM Student Members:

SL No.	USN	Name	Sem	ACM Membership ID
1	1DS22IS038	Chandana G	6	4825463
2	1DS22IS044	Disha N G	6	8481650
3	1DS22IS046	Dwaranpudi Sowmith Reddy	6	9471022
4	1DS22IS073	Lochan T N	6	1746376
5	1DS22IS077	Manaswini G	6	7585274
6	1DS22IS097	Niharika N	6	8039510
7	1DS22IS102	Pallavi P Kamath	6	9342138
8	1DS22IS108	Prakruthi S	6	0849731
9	1DS23IS021	Anurag Jain	4	3881235
10	1DS23IS030	Bharath Kumar B D	4	8288827
11	1DS23IS045	Disha T P	4	5392476
12	1DS23IS048	Gaargi L	4	7394569
13	1DS23IS051	Gautham Krishna Vijayan	4	8205740
14	1DS23IS406	Hamsa G	6	5587176

From LLM's to RAG: Enhancing AI Capabilities with External Knowledge

2024 Odd Semester

DATE: 03rd Oct 2024

ACM Faculty Sponsor: Dr. Rajeshwari J
Faculty Coordinators: Prof. Shalini Prof & Prof.
Bharath BC

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**DEPARTMENT OF INFORMATION SCIENCE & ENGINEERING**
(Accredited by NBA Tier 1: 2022-2025)

**-- DSCE-ISE ACM STUDENT CHAPTER PRESENTS --**

From LLMs to RAG: Enhancing AI Capabilities with External Knowledge

**ABHINAV**
THE CO-FOUNDER AND THE VICE
PRESIDENT OF AI AT YARNIT

ON : 03RD OCT 2024
MODE: ONLINE
TIME: 06:30 PM

Prof. Shalini K B
Prof. Bharath BC
ACM Faculty Coordinators
Asst. Professor, Dept of ISE

Dr. Rajeshwari J
ACM Faculty Sponsor
Professor,
Dept of ISE

Dr. Annapurna P Patil
Dean Academics
&HOD-ISE

Dr. B G Prasad
Principal
DSCE

Introduction:

Abhinav, Co-founder and Vice President of AI at Yarnit, conducted an insightful workshop on Retrieval Augmented Generation (RAG). The session delved into the limitations of AI models and introduced RAG as a solution to enhance their accuracy and reliability. Abhinav discussed the components of RAG and shared practical resources, providing participants with a clear understanding of this advanced AI technique. The session was conducted by the ACM Student Chapter which led to the opening of new thinking towards the path of AI.

Description:

The workshop on Retrieval Augmented Generation (RAG) conducted by Abhinav, Co-founder and Vice President of AI at Yarnit. The workshop took place on October 3rd and provided

participants with a comprehensive understanding of RAG, its importance in modern AI, and practical methods to implement it effectively.

During the workshop, Abhinav discussed the limitations of traditional AI models, such as their tendency to hallucinate or generate inaccurate responses. He explained how RAG can address these issues by retrieving relevant, up-to-date information from external sources before generating responses. Abhinav also detailed the steps involved in building a knowledge base for RAG, including source connection, parsing, chunking, embeddings, storage, and knowledge base creation. Additionally, he shared practical resources and GitHub repositories for participants to explore RAG further.

Conclusion:

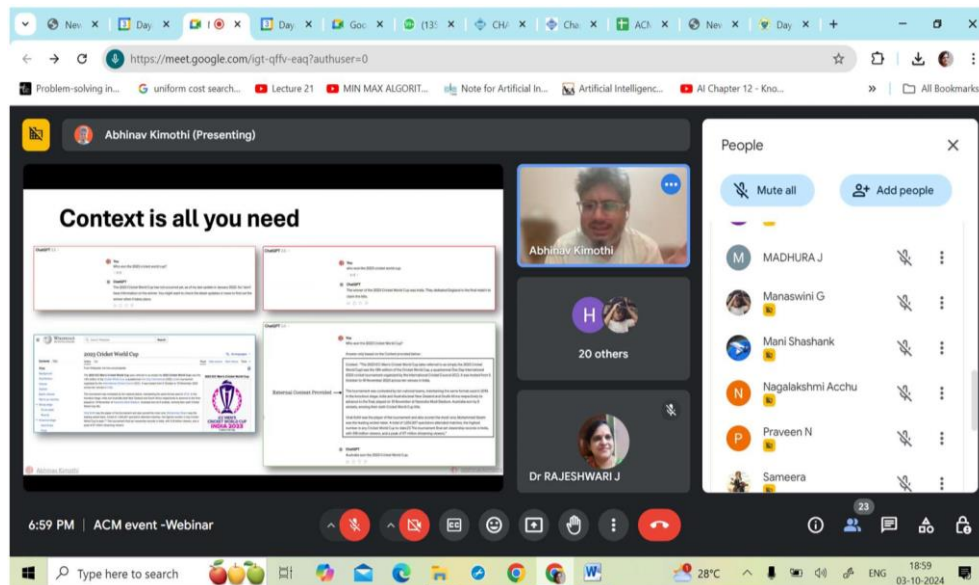
The workshop on Retrieval Augmented Generation (RAG) provided participants with a valuable understanding of this advanced AI technique. Abhinav's insightful presentation, coupled with practical resources, equipped attendees to effectively implement RAG in their own projects. Overall, the workshop was a success in enhancing participants' knowledge and skills in the field of AI.

The screenshot shows a Google Meet interface during a webinar. The main window displays a presentation slide titled "What's the benefit?". The slide is divided into two columns: "Unlimited Knowledge" and "Confidence in Responses".

Unlimited Knowledge: The text states, "The Retrieval of an AI system can have access to external sources of information. Therefore, the LLM is not limited to its internal knowledge. The external sources can be proprietary documents and data or even the internet." Below this, a diagram shows a central box labeled "Retrieval" connected to various sources like "Internal Data", "External Data", "Web", "APIs", and "Databases".

Confidence in Responses: The text states, "With the relevant source information that is retrieved made available to the LLM, the confidence in LLM responses is increased." Below this, a diagram shows a flow from "Context Awareness" to "Source Citation" to "Reduced Hallucinations".

The presentation is being presented by Abhinav Kimothi. A sidebar on the right lists participants: MADHURA J, Manaswini G, Mani Shashank, Nagalakshmi Achu, Praveen N, and Sameera. The bottom of the screen shows the Windows taskbar with the time 7:06 PM and the date 03-10-2024.



Technical Treasure Hunt

2024 Odd Semester

DATE: 07th Oct 2024

ACM Faculty Sponsor: Dr. Rajeshwari J

Faculty Cordinators: Prof. Shalini Prof, Dr. Madhura
J & Prof. Bharath BC



Introduction: The ACM Technical Treasure Hunt, a thrilling and intellectually stimulating event, was jointly organized by the ACM Student Chapter and the Genesis Student Club of the ISE Department on October 7, 2024. This event provided a unique platform for students to showcase their technical prowess and problem-solving skills in a fun and competitive environment.

Date and Time: The ACM Technical Treasure Hunt was a thrilling event held on October 7, 2024 Monday, starting at 2 PM. Participants, divided into teams, competed in solving challenging technical puzzles and riddles. The winners were awarded prizes and recognition for their exceptional performance.

Location:

The ACM Technical Treasure Hunt was held in the ISE Department of Dayananda Sagar College of Engineering on October 7, 2024. The event provided a stimulating environment for students to showcase their technical skills and problem-solving abilities.

Participants:

The ACM Technical Treasure Hunt was organized by the ACM in co-ordination with Genesis Student Club and exclusively for members of the ACM Student Chapter. The event provided a platform for members to collaborate, learn from each other, and showcase their technical skills in a fun and competitive environment.

Participants were tasked with solving challenging puzzles and riddles related to programming and computer science.

Description About the Event:

The ACM Technical Treasure Hunt was a thrilling event that challenged participants to put their technical skills to the test. Students formed teams of two members each and were tasked with solving a series of five interconnected puzzles. These puzzles covered a wide range of topics, including arrays, linked lists, stacks, binary trees, and tree traversal techniques. The event was designed to be a scavenger hunt, with each solved puzzle providing a hint that led to the location of the next puzzle. Participants had to carefully analyse the clues and apply their programming knowledge to decipher the solutions. This interactive format added an element of excitement and exploration to the competition.

Questions:

Question 1: You are a data engineer working on a system that processes batch data from a network of IoT devices. These devices send temperature readings, and your job

is to write a script that processes this data and identifies any unusual patterns. You receive a batch of N temperature readings from different IoT devices. The system runs the following algorithm on these readings to detect anomalies: 1. Multiply each reading by 7 to adjust for calibration differences. 2. For every reading, subtract 3 times the index of the reading (starting from 0). 3. Sum up the processed readings. 4. Divide the total sum by N to get the average adjusted temperature. 5. Add a safety threshold of 50 to the average. Given that the readings are [20, 30, 25, 35] and value of $N = 4$, what is the final value?

Question 2: You are developing a system that processes a singly linked list of integers. The system needs to compute a special value called the Linked List Sum, which is derived from the elements of the list based on the following algorithm: Step 1: Traverse the linked list. Step 2: For every node, if the node's position (starting from 1) is odd, add the value of the node to the total sum. Step 3: If the node's position is even, multiply the value of the node by its position and then add it to the total sum. Step 4: After traversing all nodes, add a constant value of 25 to the sum to account for overhead. Step 5: If the total number of nodes is greater than 6, subtract a penalty of 131 from the sum. You are given the following linked list: 10 -> 15 -> 8 -> 20 -> 7 -> 25 -> 30 After running this algorithm on the linked list, the Linked List Sum is N which is clue for your next hunt!

Question 3: You are given an empty stack. Perform the following operations: Push the numbers 10, 20, and 30 onto the stack. Pop the top element from the stack. Push the number 50 onto the stack. Pop the top element from the stack. Push the numbers 40 and 60 onto the stack. Finally, pop all elements from the stack, summing them as you go. What is the final sum of all the elements popped from the stack? subtract the 27 from the final sum, to get to your next clue...

Question 4: Alex and Rome are playing treasure hunting. Rome has a treasure box that Alex wants to open. Rome has N keys, each of which has a unique key number. She placed the keys in boxes numbered from 1 to N . Alex somehow cheated and came to know the key number that opens the treasure box. But he doesn't know the box number which contains the key. Alex knows that for any box 'i' the key in the i th box has a key number greater than the key numbers of the keys placed in any box j where $j < i$. Given a list of key numbers, help Alex find the box number which contains the key. Or report that Rome has not placed the correct key in any of the boxes. `indexing not 0 based array k_i ' $N = 6$ Key Numbers in Boxes: [1, 2, 3, 4, 5, 6] Key Alex is looking for: 5 N

= 4 Key Numbers in Boxes: [10, 20, 30, 40] Key Alex is looking for: 10 N = 5 Key Numbers in Boxes: [5, 10, 15, 20, 25] Key Alex is looking for: 5

Question 5: Given the inorder traversal and the preorder traversal, construct the binary tree from the traversals and the sum of all leaf nodes return the next clue inorder traversal - 40, 20, 50, 10, 60, 30, 58 preorder traversal-10, 20, 40, 50, 30, 60, 58

Winners of the ACM Technical Treasure Hunt

The ACM Technical Treasure Hunt concluded with a thrilling finish, with the top prizes going to talented teams of students. The first place was secured by the dynamic duo of Yuktha P S and Varsha H, who demonstrated exceptional problem-solving skills and teamwork throughout the competition. The second place was shared by two teams: Lochan T N and Disha N G, and Himakar M S and Vani B S, who also displayed impressive technical abilities and strategic thinking.



Tech Trek – An 8-hour Open Theme Hackathon

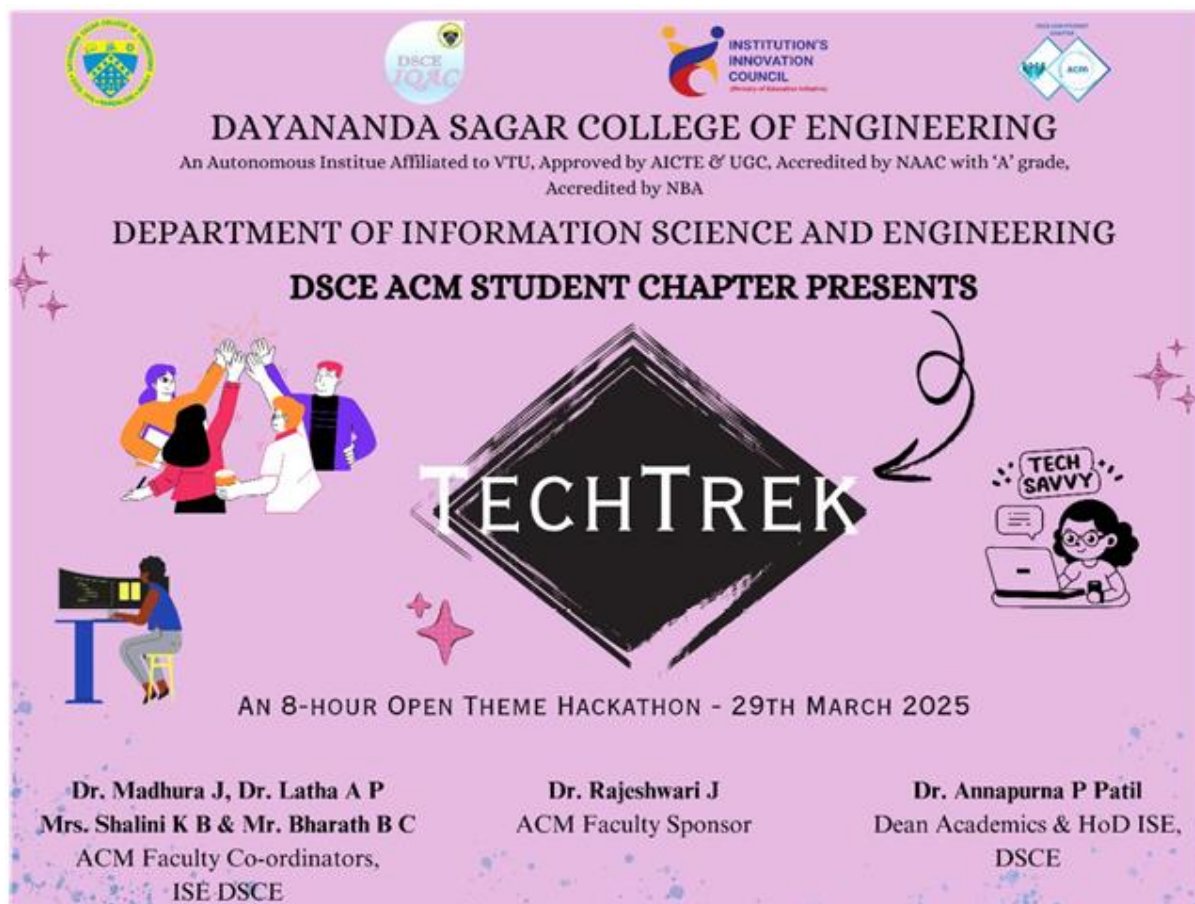
2025 Even Semester

DATE: 29th Mar 2025

ACM Faculty Sponsor: Dr. Rajeshwari J

Faculty Cordinators: Dr. Madhura J, Dr. Latha A P

Prof. Shalini K B & Prof. Bharath B C



Introduction:

The DSCE ACM Student Chapter successfully hosted TechTrek, an 8-hour open-theme hackathon, on March 29, 2025. This event provided a dynamic platform for aspiring developers and innovators to collaborate, ideate, and build creative technological solutions within a competitive yet stimulating environment. With a team size of 3-4 members, participants engaged in intensive problem-solving, leveraging cutting-edge technologies to develop impactful projects. TechTrek fostered innovation, teamwork, and technical excellence, reinforcing DSCE ACM's commitment to nurturing talent and promoting a culture of continuous learning in the field of computer science and engineering.

Date and Time:

TechTrek was conducted on March 29, 2025, as a rigorous 8-hour hackathon, running from 9:00 AM to 5:00 PM. The event was structured to provide participants with a focused and immersive coding experience within this time frame, encouraging rapid ideation, development, and execution of innovative solutions. The well-defined schedule ensured a seamless flow of activities, including project brainstorming, coding, mentoring sessions, and final presentations, culminating in the evaluation and recognition of outstanding projects.

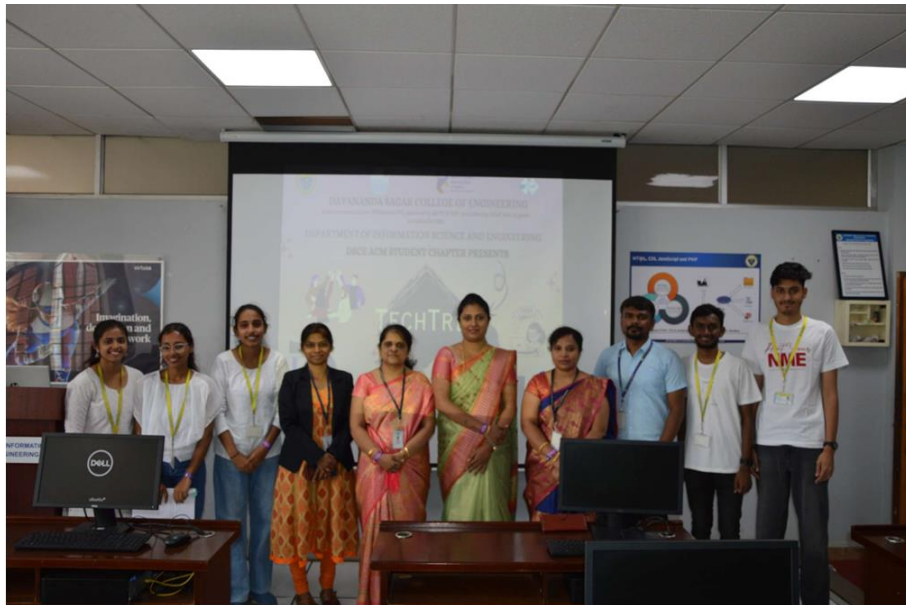
Description:

TechTrek was an 8-hour open-theme hackathon organized by the DSCE ACM Student Chapter on March 29, 2025, providing a platform for aspiring developers and problem solvers to collaborate and create innovative technological solutions. With a team size of 3-4 members, participants were encouraged to work on projects of their choice, addressing real-world challenges using cutting-edge technologies. The event fostered an environment of creativity, teamwork, and technical excellence, allowing students to showcase their skills while gaining valuable insights from mentors and industry professionals. The hackathon followed a structured format, beginning with an inaugural session and transitioning into an intensive coding phase, where teams developed their solutions under expert guidance. Midway checkpoints ensured steady progress, with mentors offering constructive feedback to help refine ideas. The event concluded with final project presentations, judged by Ms. Nishika Tiku and Dr. Mary Cherian, a professor from the ISE Department. Their expertise in industry and academia provided a well-rounded evaluation of the projects based on innovation, feasibility, and technical execution. TechTrek successfully nurtured a culture of learning and innovation, reinforcing DSCE ACM's mission to empower students through technology-driven initiatives.

Conclusion:

TechTrek concluded as a resounding success, fostering innovation, collaboration, and technical excellence among participants. The event provided a dynamic platform for students to apply their skills, develop creative solutions, and gain valuable insights from industry and academic experts. The guidance from mentors and the expert evaluation by judges, Ms. Nishika Tiku and Dr. Mary Cheiran, ensured a high standard of competition and learning. By encouraging problem-solving, teamwork, and hands-on experience with emerging technologies, TechTrek reinforced DSCE ACM's commitment to nurturing talent and promoting a culture of continuous learning and innovation in the field of computer science and engineering.





CypherQuest 2k25 - 24 Hours ACM Zonal Level Hackathon
 2025 Even Semester ACM Faculty Sponsor: Dr. Rajeshwari J
 DATE: 25th & 26th Faculty Coordinators: Dr. Madhura J, Prof. Shalini K B
 April 2025 & Prof. Bharath B C



**DAYANANDA SAGAR ACADEMY OF TECHNOLOGY
AND MANAGEMENT**

(DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)
IN ASSOCIATION WITH



DAYANANDA SAGAR COLLEGE OF ENGINEERING
(DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING)



x



presents



CYPHERQUEST 2K25

Think | Create | Innovate

**24 HOURS ACM ZONAL LEVEL
HACKATHON**



**PRICE POOL
50,000**



TO REGISTER



DAYANANDA SAGAR ACADEMY OF TECHNOLOGY AND
MANAGEMENT



&



+91 9448732789
+91 8431052327



almatron@dsatm.edu.in

Introduction:

In an initiative to cultivate innovation, collaboration, and practical technical skills among young minds, CypherQuest 2K25 – a 24-hour ACM Zonal Level Hackathon – was

organized with great enthusiasm by the DSCE ACM Student Chapter and DSATM ACM Student Chapter.

Held in collaboration with Dayananda Sagar College of Engineering (DSCE) and Dayananda Sagar Academy of Technology and Management (DSATM), this grand event took place on April 25–26, 2025, offering a vibrant platform for students to brainstorm, build, and showcase their creative solutions to real-world challenges.

Participation Overview:

CypherQuest 2K25 witnessed an overwhelming response with 210 participants grouped into 55 teams. Students from various reputed institutions across Karnataka and beyond participated, including:

Dayananda Sagar College of Engineering (DSCE), Dayananda Sagar Academy of Technology and Management (DSATM), Dayananda Sagar University (DSU), Sri Venkateshwara College of Engineering, Cambridge Institute of Technology, Bengaluru, T John Institute of Technology, The Oxford College of Engineering, Bengaluru, New Horizon College of Engineering, SJB Institute of Technology, RV Institute of Technology and Management, RV College of Engineering, Acharya Institute of Technology, JSS Academy of Technical Education, Bengaluru, Nitte Meenakshi Institute of Technology, KLE Technological University (formerly BVB CET), Hubballi, Kalpataru Institute of Technology, Tiptur, Don Bosco Institute of Technology, Bengaluru.

The teams showcased immense diversity in their approach, backgrounds, and technical domains, ranging from AI/ML and Blockchain to Web Development and Sustainable Tech Solutions. Hackathon Environment and Facilities: The hackathon environment was meticulously designed to ensure high productivity, learning, and comfort for participants:

- **Healthy and Timely Meals:** Dinner and Breakfast arrangements ensured that participants remained energized throughout the event.
- **Rest Areas:** Lounge rooms were arranged, providing participants a space to relax, recharge, and continue their work with renewed energy.
- **Technical Support:** Continuous access to Wi-Fi, on-site technical mentors, and immediate troubleshooting teams allowed participants to stay focused on innovation.
- **Safe and Secure Venue:** 24/7 security arrangements and medical facilities were made available to prioritize the well-being of all attendees. The spirit throughout the

hackathon was electric, with participants coding intensely, brainstorming collaboratively, and overcoming challenges with resilience.

Timeline and Structure:

The event followed a well-organized timeline:

- 9:30 AM: Check-in and Registration
- 10:30 AM: Inauguration Ceremony
- 12:00 PM: Lunch Break
- 12:30 PM: Lab Allotment
- 1:00 PM: Hackathon Kick-off
- 9:00 PM: Dinner Break
- Overnight: Continuous Coding and Development; Lounge Room Access
- 9:00 AM (Next Day): Breakfast and Final Touches
- 1:00 PM: Evaluation Phase and Project Demonstrations
- Following Evaluation: Announcement of Results and Closing Ceremony

Judging and Evaluation:

The projects were evaluated by an eminent panel of judges, alumni of DSCE, who are currently excelling in their respective fields. The judges included:

- Akash HR, Senior HPC Infrastructure and DevOps Engineer (Software Engineer 2) at SanDisk India Device Design Centre Pvt. Ltd.
- Dinesh A, Software Development Engineer (SDE-1) at Simple Energy.
- Abdul Rawoof Khan, Software Development Engineer 2 at IQVIA.
- Preethik Shetty, who also brought his rich experience to the evaluation process.

Their technical expertise and critical evaluation helped identify the most innovative, impactful, and feasible projects.

Outcomes and Benefits:

The CypherQuest 2K25 hackathon contributed significantly to the United Nations Sustainable Development Goals (SDGs):

- Skill Development – Improved technical and problem-solving abilities. → Mapped to SDG 4: Quality Education

- Innovation Promotion – Fostered creativity and technological innovation. → Mapped to SDG 9: Industry, Innovation, and Infrastructure
- Real-World Problem Solving – Addressed challenges in health, environment, and urban development. → Mapped to SDG 11: Sustainable Cities and Communities → Linked to SDG 3: Good Health and Well-Being
- Teamwork and Collaboration – Strengthened teamwork, partnerships, and knowledge sharing. → Mapped to SDG 17: Partnerships for the Goals
- Responsible Innovation – Promoted ethical and sustainable development practices. → Mapped to SDG 12: Responsible Consumption and Production
- Inclusive Participation – Offered equal opportunity to students from diverse backgrounds. → Mapped to SDG 10: Reduced Inequalities

Conclusion:

CypherQuest 2K25 was a resounding success — a celebration of technology, teamwork, and tenacity. It reflected the values of innovation, leadership, and collaboration nurtured within the DSCE and DSATM communities. The hackathon not only unleashed the creative potential of the participants but also fostered connections, built lifelong memories, and set a high benchmark for future events.





Developers' Conclave: Industry Driven Insights from Industry Insiders

2025 Even Semester

ACM Faculty Sponsor: Dr. Rajeshwari J

DATE: 26th April
2025

Head of the Department: Dr. Vindhya P Malagi

Faculty Cordinators: Dr. Reshma S

Artificial Intelligence & Machine Learning, DSCE

Purpose:

The Developer Conclave on Design & Analysis of Algorithms (DAA) is a focused initiative for 4th-semester AI & ML students, aimed at bridging the gap between academic learning and industry requirements. In today's technology-driven world, algorithmic efficiency forms the backbone of AI, data science, and software development. This event is designed to transform theoretical knowledge into practical skills, preparing students for technical careers while emphasizing the real-world impact of optimized algorithms.

Conduction:

- Scheduled for 26th April 2025, the conclave will feature:
- Industry Expert Sessions: 14 professionals from leading IT companies will share insights on algorithmic applications in real-world scenarios.
- Panel Discussions: Open forums on placement strategies, interview preparation, and emerging trends in algorithms.
- Q&A Sessions: Direct engagement with experts for career guidance and doubt resolution.

Benefits to Participants:

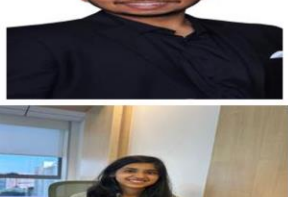
- Gain industry-relevant perspectives on algorithm design and optimization.
- Learn placement strategies and technical interview techniques from experts.
- Understand the role of algorithms in AI, machine learning, and large-scale systems.
- Network with professionals and peers to expand career opportunities.
- Receive guidance on continuous skill development to stay competitive in the field.

The Developer Conclave serves as a vital platform for students to align their academic knowledge with industry demands. By focusing on algorithmic thinking, coding proficiency, and career readiness, the event empowers participants to tackle complex challenges with confidence. The insights gained will not only enhance their technical

capabilities but also inspire a mindset of innovation and continuous learning. As algorithms continue to drive technological advancements, this conclave will equip students to become proficient, industry-ready professionals.

About The Developers

Sl No	Developer Name	Company Name	Designation	Photo
1	Bhagyashree bhjayara@in.ibm.com +91-9880714498	IBM 26 Year	Executive IT Architect (L3 Certified) , Distinguished Architect	
2	Prakash Narkodu prakash.narkodu@hcltech.com	HCL Tech 23 Years	Technical Specialist	
3	Priyanka Raghavan	A.P. Moller – Maersk 20 Years	Cyber Lead/Senior Cyber Manager@ A.P. Moller - Maersk Cybers ecurity Strategy and Architecture	
4	Harsha Chakravarthy K.N Harsha.neella@wipro.com	Wipro Limited 16 Years	Privacy Governance Head, Global Data Privacy	
5	Sridhar Hegde	Rockwell automations 15 Years	Industrialization manager	
6	Ranganidhi V J	CGI Services 14 Years	Business Analyst	

7	Abhijith S	Skyhigh Security 11 Years	Technical lead in Skyhigh	
8	Gopika P	Emirates 8 Years	Senior software engineer	
9	Nithin V Gopal	Amagi Media Labs 7 Years	Product Development Lead	
10	Bhavishya Pandit	IBM 4 Years	AI Engineer	
11	Ashutosh Pandey	TEKsystems Global Services in India 4 Years	Senior Compiler Engineer	
12	Sumedh M Anvekar	TEKsystems Global Services in India 2 Years	Associate Software Engineer	
13	C H NagaNoushith		Associate software Engg	
14	Samiksha	Alcon laboratories pvt ltd 1 Year	Data Analyst	

Feedback And Suggestion from Developer

The feedback from the developer emphasized the need for breakout sessions tailored to specific interests. These sessions would be highly beneficial for attendees by allowing them to delve deeper into their areas of interest and gain valuable knowledge related to their chosen fields.

OUTCOMES

The outcome of the developer talk focused on several key takeaways for aspiring developers:

- **Industry Exposure:** Students and attendees gained direct insights from experienced professionals across leading tech companies like IBM, Wipro, AMD, HCL, and more.
- **Career Guidance:** Panel discussions provided valuable advice on career paths, skill development, and industry expectations in the tech and automation sectors.
- **Networking Opportunities:** Participants had the chance to connect with industry experts, potentially opening doors for internships, mentorships, or future collaborations.
- **Awareness of Trends:** The sessions highlighted current trends in automation, digital transformation, cybersecurity, and smart manufacturing.
- **Inspiration and Motivation:** Real-life experiences and success stories shared by the speakers served to inspire students and young developers.





Aventus 3.0 - 24 hrs National Level Hackathon

2025 Even Semester

DATE: 17th & 18th
May 2025

ACM Faculty Sponsor: Dr. Rajeshwari J
Head of the Department: Dr. Vindhya P Malagi
Faculty Cordinators: Prof. Kavya D N
Artificial Intelligence & Machine Learning, DSCE



The banner for Aventus 3.0 features a dark blue background with a network of white dots and lines. At the top, logos for DSCE IQAC, Institution's Innovation Council, Dayananda Sagar College of Engineering, and ACM are displayed. The central text reads: "Dayananda Sagar College of Engineering", "Innovation Challenge Series under the aegis of DSCE Innovation Council & ACM Student Chapter", "Department of Artificial Intelligence and Machine Learning", "In Collaboration with", and "Techmiya Solutions and DERBI Foundation". Below this, logos for Tech for People, DERBI Foundation, Sagar Hospitals, and mentorlink are shown. The main title "AVENTUS 3.0" is prominently displayed in a stylized font, followed by "A National Level Hackathon" and "May 17th - 18th, 2025". At the bottom, four icons represent the event's features: ₹3 Lakh Prize Bounty, 24 Hours Non-Stop Coding!, Innovation Solution to Real-World Challenges, and Incubation Opportunities!

DSCE IQAC

INSTITUTION'S INNOVATION COUNCIL

DAYANANDA SAGAR COLLEGE OF ENGINEERING

acm

Dayananda Sagar College of Engineering
Innovation Challenge Series under the aegis of DSCE Innovation Council & ACM Student Chapter
Department of Artificial Intelligence and Machine Learning
In Collaboration with
Techmiya Solutions and DERBI Foundation

Tech for People

DERBI[®]
FOUNDATION
...Fueling Dreams

Sagar Hospitals[®]
With You. For Life.

AVENTUS 3.0
A National Level Hackathon
May 17th - 18th, 2025

mentorlink

₹3 Lakh
Prize Bounty

24 Hours
Non-Stop Coding!

Innovation Solution to
Real-World Challenges

Incubation
Opportunities!

Objective: The event was organized with the vision of nurturing a culture of innovation, problem-solving and technical excellence among young minds. It aimed to provide a national platform where students could channel their creativity and technical knowledge to develop practical solutions to pressing real-world issues. By bringing together participants from diverse academic backgrounds, the goal was to encourage

cross-disciplinary collaboration, hands-on learning and exposure to the latest technological trends.

Description: The 24-hour on-campus hackathon brought together talented student developers from across the country to collaborate, innovate, and build impactful tech solutions. Organized by the Department of Artificial Intelligence and Machine Learning in association with the ACM Student Chapter and the Institution's Innovation Council (IIC), the event provided an energetic and immersive environment where participants tackled real-world challenges and developed functional prototypes within a limited timeframe. With continuous mentor support and expert-led sessions offering insights into industry trends and emerging technologies, teams refined their solutions through multiple evaluation rounds. The hackathon concluded with a grand prize distribution and valedictory ceremony, celebrating innovation, collaboration and excellence in problem-solving.

The hackathon was inaugurated on May 17, 2025, at 9:30 AM with an inspiring opening ceremony. Esteemed guests—Ms. Revathi Kamath, Philanthropist & Entrepreneur; Dr. Raja Dutta, Medical Director, Sagar Hospitals; Mr. Mohammad Yasin Shabhaz, CEO, Techmiya Solutions; Mr. Vivek Singhal, Co-founder & Chief Data Scientist, Cellstrat AI Lab; Mr. Sathyanarayana B V, CEO, DERBI Foundation, DSI; and Dr. B. G. Prasad, Principal, Dayananda Sagar College of Engineering, shared visionary insights that set an electrifying tone for the mega event. By 11:00 AM, the 24-hour innovation sprint was officially underway, with teams diving into bold, real-world problem-solving.

Mentors with expertise across diverse technical and business domains were assigned to each team, ensuring personalized guidance throughout the hackathon. Three structured mentoring rounds were conducted at key intervals, allowing teams to present their progress, receive critical feedback and pivot or refine their ideas as needed. Beyond these rounds, mentors were available throughout the night providing ongoing support and encouragement to help teams overcome challenges and push their projects toward impactful outcomes.

During the final round of Aventus 3.0, the 12 shortlisted teams took the stage at the PC Sagar Auditorium to present their projects to the grand juries. Each team delivered a focused presentation followed by a Q&A session, where they defended their ideas and demonstrated technical depth. Following intense deliberation, the grand jury concluded the event with a valedictory ceremony and prize distribution at 3:00 PM.

- **Team Code Krafters** from **Bangalore Institute of Technology** claimed **1st place**, winning a cash prize of **₹1,00,000**.
- **Team I/O** from **RV College of Engineering** secured **2nd place**, receiving **₹50,000**, and
- **Team Marinira Saas** from **Dayananda Sagar College of Engineering** earned **3rd place**, with **₹30,000**.

Bounty prizes of ₹10,000 each were awarded to:

- **Team Zephyr** from **M S Ramaiah Institute of Technology** for the **Most Innovative Solution**,
- **Team Fem Tech** from **Dayananda Sagar College of Engineering** for **promoting Women in Tech** and
- **Team Game Changers** from **Dayananda Sagar College of Engineering** for addressing **Sustainable Development Goals**.

In addition, **DERBI Foundation** recognized exceptional business potential by awarding **₹25,000 each** to **Team Monkey Code** and **Team Charminar** for the **Best Business Solutions** from **Dayananda Sagar College of Engineering**. The ceremony marked a powerful conclusion to Aventus 3.0 by celebrating innovation, inclusion and entrepreneurial excellence.

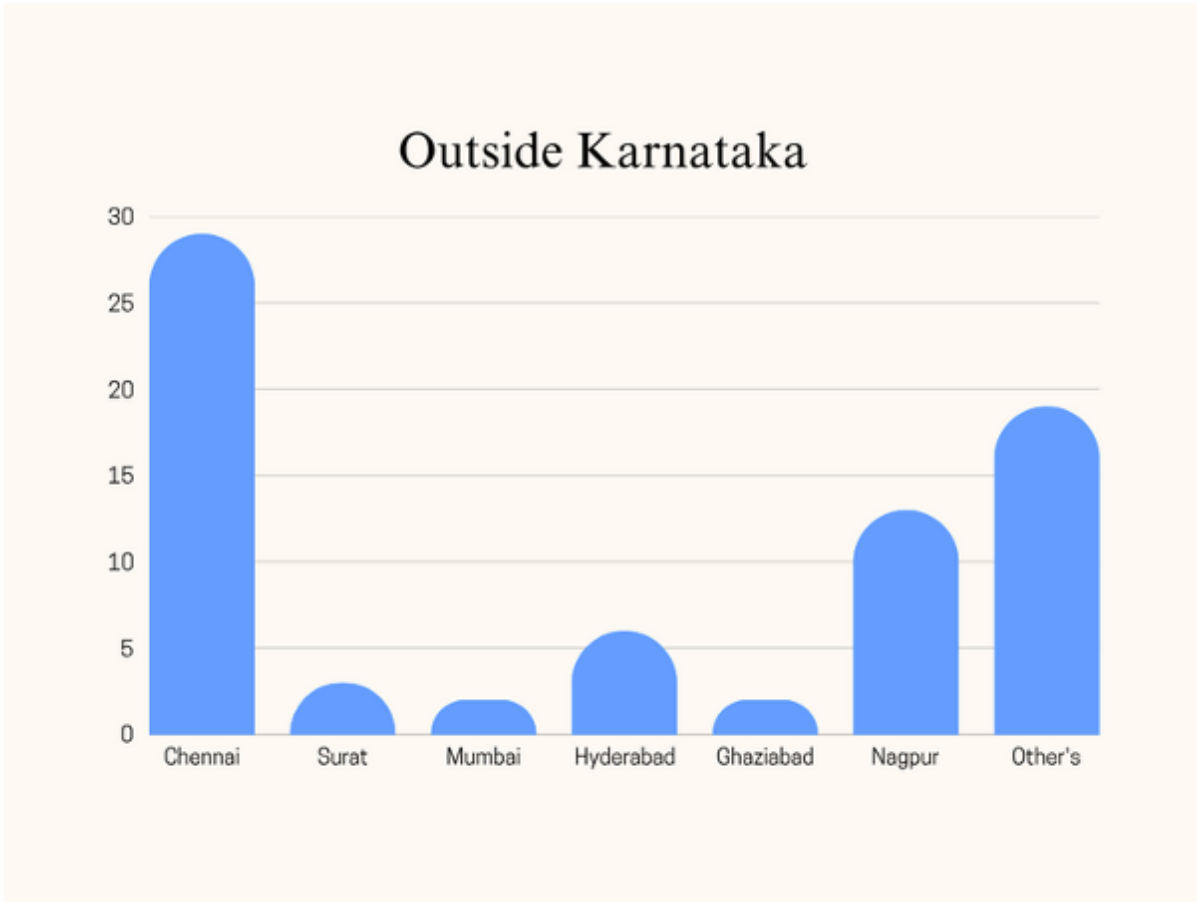
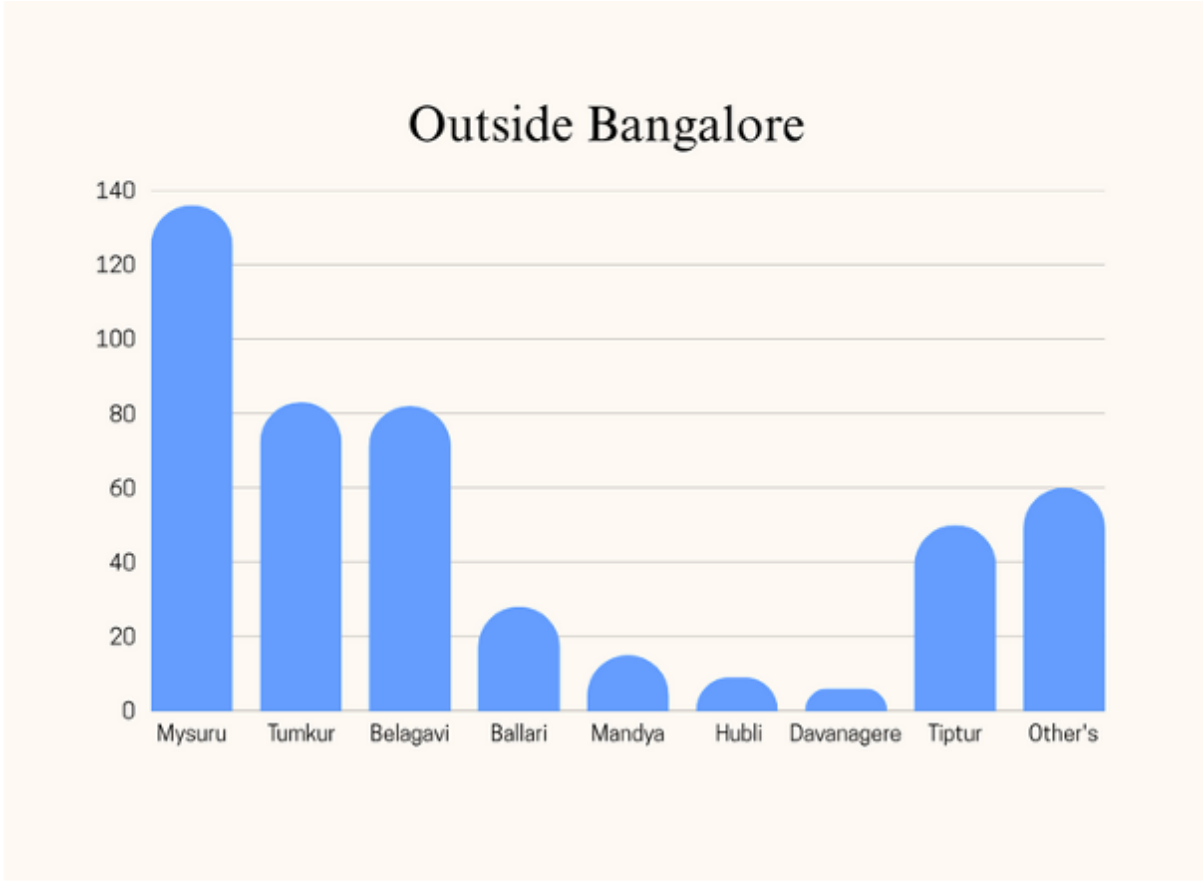
Judges:

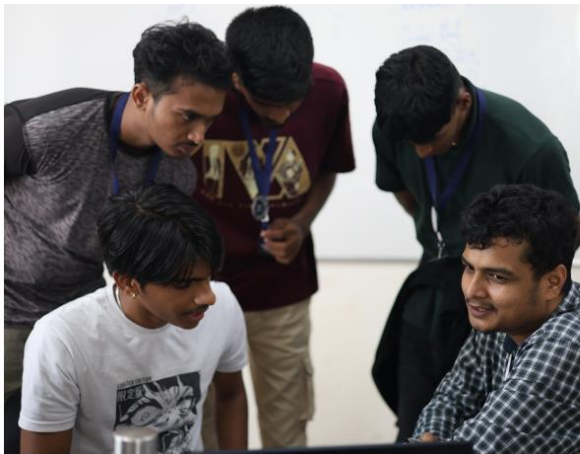
Joel	AI in Healthcare
Samarth	AI in Healthcare
Naman	AI in CyberSecurity
Adithya	AI in CyberSecurity
Rishab	AI in CyberSecurity
Shishi Prakash	AI in Edge Computing
Ashutosh	AI in Edge Computing
Nirmal	AI in Defence & Disaster Management
Hanu	AI in Defence & Disaster Management
Bapu	AI in Defence & Disaster Management

Grand Jury:

Mr. Mandar M	Engineering Leadership, Amazon
Mr. Naveen Namperumal	Delivery Head, V2Soft
Mr. Mohammed Yasin Shabaz	CEO, Techmiya Solutions
Mr. Sathyanarayana B V	CEO, DERBI Foundation

Regional Diversity of Participants:









AI-assisted Tutoring for CS1

2025 Even Semester

ACM Faculty Sponsor: Dr. Rajeshwari J

DATE: 27th June 2025

Faculty Coordinators: Dr. Prathima Mabel, Dr. Vani
KA & Prof. Bharath B C

DAYANANDA SAGAR COLLEGE OF ENGINEERING

An Autonomous Institute Affiliated to VTU,

Approved by AICTE & UGC, Accredited by NAAC with 'A' grade, & ISO 9001-2015 Certified Institution

DEPARTMENT OF INFORMATION SCIENCE & ENGINEERING

(ACCREDITED BY NBA TIER 1: 2025-2028)



--DSCE ACM STUDENT CHAPTER PRESENTS-- TECHNICAL TALK ON AI ASSISTED TUTORING FOR CS1

DR. AMEY KARKARE

POONAM AND PRABHU GOEL CHAIR PROFESSOR,
DEPT OF CSE
IIT KANPUR



27 JUNE, 2025



11:30 AM



ISE 308, DSCE

DR. PRATHIMA MABEL
DR. VANI KA
PROF. BHARATH BC
FACULTY COORDINATORS

DR. RAJESHWARI J
ACM FACULTY SPONSOR
PROFESSOR, ISE

DR. ANNAPURNA P PATIL
DEAN ACADEMICS
& HOD-ISE

DR. B G PRASAD
PRINCIPAL
DSCE

Objective

To explore the integration of AI-assisted tutoring tools in the CS1 curriculum to enhance student learning, engagement, and problem-solving skills.

Overview

The session focused on how Artificial Intelligence (AI) can support beginners in Computer Science (CS1) by providing personalized guidance, instant feedback, and adaptive learning paths. It highlighted current AI tutoring technologies, their benefits, and challenges in implementation.

Key Points Discussed

- **Personalized Learning:** AI tutors can adapt to individual student pace and style, offering customized exercises and explanations.
- **Instant Feedback:** Automated code analysis and error detection help students identify and correct mistakes quickly.
- **Interactive Problem Solving:** AI tools can simulate real-time coding environments and guide students through complex concepts.
- **Data-Driven Insights:** Tracking student progress and common errors to tailor future lessons.
- **Limitations:** AI cannot fully replace human instructors; ethical considerations and dependency risks were discussed.
- **Examples of Tools:** Platforms like CodeCoach, Mimir Classroom, and AI chatbots for coding assistance.

Outcomes

- Students expressed increased confidence in tackling programming assignments.
- Recognition of AI tutoring as a valuable supplement rather than a

replacement for traditional teaching.

- Identification of areas needing further research, such as improving AI's contextual understanding and emotional support.
- Agreement on the potential of AI to reduce dropout rates by providing timely support.

Conclusion

AI-assisted tutoring presents a promising avenue to enhance CS1 education by offering personalized, immediate, and scalable support to students. While it cannot replace human interaction, its integration can significantly improve learning outcomes and student motivation when used thoughtfully alongside traditional teaching methods.

